

# Search log

One row per search run. Add a row every time a string is run in a database, and paste the string exactly as it was run in the section below.

| Date       | Database                                 | Interface                                                          | ID | Fields          | Limits                                                                    | Hits               | Export                                    | Run by                           |
|------------|------------------------------------------|--------------------------------------------------------------------|----|-----------------|---------------------------------------------------------------------------|--------------------|-------------------------------------------|----------------------------------|
| 2026-10-06 | arXiv                                    | API, export.arxiv.org/api/query                                    | S0 | title, abstract | submitted 2020-01-01 to 2026-10-06                                        | 3                  | exports/arxiv/2026-10-06/raw/S0_*.xml     | Saad, scripts/arxiv_search.py    |
| 2026-10-06 | arXiv                                    | API                                                                | S1 | title, abstract | same                                                                      | 270                | .../raw/S1_*.xml                          | same                             |
| 2026-10-06 | arXiv                                    | API                                                                | S2 | title, abstract | same                                                                      | 474                | .../raw/S2_*.xml                          | same                             |
| 2026-10-06 | arXiv                                    | API                                                                | S3 | title, abstract | same                                                                      | 945                | .../raw/S3_*.xml                          | same                             |
| 2026-10-06 | ACL Anthology                            | Local filter of anthology+abstracts.bib.gz (file dated 2026-10-05) | S0 | title, abstract | year 2020 or later; conference papers and journal articles                | 1                  | exports/acl/2026-10-06/S0.bib             | Saad, scripts/acl_filter.py      |
| 2026-10-06 | ACL Anthology                            | Local filter                                                       | S1 | title, abstract | same                                                                      | 40                 | .../S1.bib                                | same                             |
| 2026-10-06 | ACL Anthology                            | Local filter                                                       | S2 | title, abstract | same                                                                      | 90                 | .../S2.bib                                | same                             |
| 2026-10-06 | ACL Anthology                            | Local filter                                                       | S3 | title, abstract | same                                                                      | 151                | .../S3.bib                                | same                             |
| 2026-10-06 | OpenAlex, IEEE- and ACM-published papers | API, api.openalex.org/works                                        | S0 | title, abstract | published 2020-01-01 or later; DOI prefix 10.1109 (IEEE) or 10.1145 (ACM) | IEEE 1 · ACM 1     | exports/openalex/2026-10-06/raw/S0_*.json | Saad, scripts/openalex_search.py |
| 2026-10-06 | OpenAlex, IEEE and ACM                   | API                                                                | S1 | title, abstract | same                                                                      | IEEE 176 · ACM 54  | .../raw/S1_*.json                         | same                             |
| 2026-10-06 | OpenAlex, IEEE and ACM                   | API                                                                | S2 | title, abstract | same                                                                      | IEEE 91 · ACM 51   | .../raw/S2_*.json                         | same                             |
| 2026-10-06 | OpenAlex, IEEE and ACM                   | API                                                                | S3 | title, abstract | same                                                                      | IEEE 508 · ACM 195 | .../raw/S3_*.json                         | same                             |

## Totals so far

|                                                                | Records |
|----------------------------------------------------------------|---------|
| Identified on arXiv (four strings)                             | 1,692   |
| Identified in the ACL Anthology (four strings)                 | 282     |
| Identified in OpenAlex, IEEE- and ACM-published (four strings) | 1,077   |

|                                       | Records      |
|---------------------------------------|--------------|
| <b>Identified in total</b>            | <b>3,051</b> |
| Duplicates removed (table at the end) | 496          |
| <b>Unique records to screen</b>       | <b>2,555</b> |

Not run: Scopus, and IEEE Xplore and the ACM Digital Library themselves. IEEE and ACM publications were reached through OpenAlex instead; the limits of that are in the notes on the OpenAlex run.

## Strings exactly as run

### arXiv, 2026-10-06

S0

```
(ti:"retrieval augmented" OR abs:"retrieval augmented" OR ti:RAG OR abs:RAG) AND (ti:email OR abs:email OR ti:emails OR abs:emails OR ti:"e-mail" OR abs:"e-mail" OR ti:"e-mails" OR abs:"e-mails") AND (ti:attachment OR abs:attachment OR ti:attachments OR abs:attachments OR ti:multimodal OR abs:multimodal OR ti:"multi-modal" OR abs:"multi-modal") AND submittedDate:[202001010000 TO 202610062359]
```

S1

```
(ti:"retrieval augmented" OR abs:"retrieval augmented" OR ti:RAG OR abs:RAG) AND (ti:email OR abs:email OR ti:emails OR abs:emails OR ti:"e-mail" OR abs:"e-mail" OR ti:"e-mails" OR abs:"e-mails" OR ti:mailbox OR abs:mailbox OR ti:inbox OR abs:inbox OR ti:enterprise OR abs:enterprise OR ti:"company-internal" OR abs:"company-internal" OR ti:workplace OR abs:workplace) AND submittedDate:[202001010000 TO 202610062359]
```

S2

```
(ti:"retrieval augmented" OR abs:"retrieval augmented" OR ti:RAG OR abs:RAG) AND (ti:"knowledge conflict" OR abs:"knowledge conflict" OR ti:"knowledge conflicts" OR abs:"knowledge conflicts" OR ti:"conflicting evidence" OR abs:"conflicting evidence" OR ti:deduplication OR abs:deduplication OR ti:"near-duplicate" OR abs:"near-duplicate" OR ti:redundancy OR abs:redundancy OR ti:outdated OR abs:outdated OR ti:superseded OR abs:superseded) AND submittedDate:[202001010000 TO 202610062359]
```

S3

```
(ti:"retrieval augmented" OR abs:"retrieval augmented" OR ti:RAG OR abs:RAG) AND (ti:multimodal OR abs:multimodal OR ti:"multi-modal" OR abs:"multi-modal" OR ti:"visual document" OR abs:"visual document" OR ti:OCR OR abs:OCR) AND submittedDate:[202001010000 TO 202610062359]
```

### ACL Anthology, 2026-10-06

The Anthology has no Boolean search, so its full bibliography file was filtered locally with the same blocks. The terms of each block are those of scripts/search\_blocks.py, identical to the arXiv strings above.

```
S0 = RAG AND EMAIL AND ATTACHMENT
S1 = RAG AND CONTEXT
S2 = RAG AND HYGIENE
S3 = RAG AND MODALITY
```

- **Source file:** <https://aclanthology.org/anthology+abstracts.bib.gz>, last modified 2026-10-05, 131,647 entries, SHA-256 d9620aa9651f86bfb5177b3825adda20b18621239a696062d15c8befb1c7573e.
- **Entries considered:** 68,810 conference papers and journal articles dated 2020 or later.
- **Matching rule:** a term matches in the title or abstract, case-insensitive, as whole words, with hyphen and space treated alike and an optional plural "s".

## OpenAlex, 2026-10-06

Each string was run twice, once per DOI prefix, as the filter parameter:

```
title_and_abstract.search:<string>,from_publication_date:2020-01-01,doi_starts_with:10.1109  
title_and_abstract.search:<string>,from_publication_date:2020-01-01,doi_starts_with:10.1145
```

S0

```
("retrieval augmented" OR RAG) AND (email OR emails OR "e-mail" OR "e-mails") AND (attachment OR attachments OR  
multimodal OR "multi-modal")
```

S1

```
("retrieval augmented" OR RAG) AND (email OR emails OR "e-mail" OR "e-mails" OR mailbox OR inbox OR enterprise OR  
"company-internal" OR workplace)
```

S2

```
("retrieval augmented" OR RAG) AND ("knowledge conflict" OR "knowledge conflicts" OR "conflicting evidence" OR  
deduplication OR "near-duplicate" OR redundancy OR outdated OR superseded)
```

S3

```
("retrieval augmented" OR RAG) AND (multimodal OR "multi-modal" OR "visual document" OR OCR)
```

## Notes on the arXiv run

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- **Seed check.** 14 of the 21 proposal references indexed on arXiv are retrieved by at least one string. Not retrieved: [1], [6], [7], [9], [15], [22], [23]. Details per reference are in `run.json`.
- **S1 by submission year.** 2021: 1 · 2023: 1 · 2024: 40 · 2025: 94 · 2026: 134.
- **Categories.** 1,583 of the 1,612 unique records have a `cs.*` primary category. The other 29 should be checked at screening for other meanings of the acronym RAG.
- **Gap check (S0).** Three records: arXiv 2504.13209, 2509.11937 (MMORE) and 2603.01990 (ATM-Bench).

## Notes on the ACL Anthology run

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- **Unique records.** 273 across the four strings. Two of them have no abstract in the source and were matched on the title alone.
- **By year.** 2021: 2 · 2022: 3 · 2023: 5 · 2024: 42 · 2025: 114 · 2026: 107.
- **Overlap with arXiv.** 174 of the 273 have the same title and at least one author in common with a record of the arXiv run. They are listed in `exports/acl/2026-10-06/run.json`.
- **Gap check (S0).** One record: 2026.1an1p-1.6 (mCS-LM), not on arXiv.

## Notes on the OpenAlex run

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- **What this run is.** A search of OpenAlex, an open index of scholarly works, limited to DOIs registered by IEEE and ACM. It is not a search of IEEE Xplore or of the ACM Digital Library, which cannot be queried by script.
- **Unique records.** 1,034 across the four strings: 749 published by IEEE and 285 by ACM.
- **Missing abstracts.** 64 of the 1,034 have no abstract in OpenAlex (56 IEEE, 8 ACM) and matched on the title alone. Papers whose abstract is missing in OpenAlex and whose title lacks the terms are not retrieved; their number is unknown.

- **One DOI indexed twice.** For S3 on ACM, OpenAlex returns 195 works, two of which carry the same DOI (10.1145/3811238.3811871). They count as one record.
- **By year.** 2021: 1 · 2022: 2 · 2023: 7 · 2024: 95 · 2025: 530 · 2026: 399.
- **Overlap.** 190 of the 1,034 have the same title and at least one author in common with an arXiv record. They are listed in `exports/openalex/2026-10-06/run.json`.
- **Non-papers.** About a dozen records are editorials, conference abstracts, reviews or front matter, to exclude at screening.
- **Gap check (S0).** Two records: 10.1109/ickecs70176.2026.11528041 (AutoMeet) and 10.1145/3805712.3808647.

## Loaded into Zotero

The shared library for the review is the Zotero group **FrAnderl-RAG-Review**.

| Date       | What                            | Records                       | Tags                                                                                 |
|------------|---------------------------------|-------------------------------|--------------------------------------------------------------------------------------|
| 2026-10-06 | arXiv S0 and S1                 | 270                           | search:S0, search:S1, db:arxiv, run:2026-10-06                                       |
| 2026-10-06 | Proposal references             | 24, of which 5 are also in S1 | seed                                                                                 |
| 2026-10-06 | arXiv S2 and S3                 | 1,333 new                     | search:S2, search:S3, db:arxiv, run:2026-10-06                                       |
| 2026-10-06 | ACL Anthology S0 to S3          | 99 new                        | search:S0 to search:S3, db:ac1, run:2026-10-06                                       |
| 2026-10-06 | OpenAlex S0 to S3, IEEE and ACM | 844 new                       | search:S0 to search:S3, db:openalex, publisher:ieee or publisher:acm, run:2026-10-06 |

- **Total loaded by script:** 2,565 items. That is the 2,555 unique records, plus the 10 proposal references that no string retrieves.
- **OpenAlex:** the 190 records that duplicate an arXiv record were not loaded. For those papers the arXiv record is the one in Zotero, and the published metadata is in `exports/openalex/2026-10-06/records.json`.
- **arXiv S2 and S3:** 1,342 unique records lie outside S1. Nine of them were already in the library as proposal references, so 1,333 were added.
- **ACL Anthology:** the 174 records that duplicate an arXiv record were not loaded. For those papers the arXiv record is the one in Zotero, and the published metadata is in `exports/ac1/2026-10-06/records.json`.
- **Not from a logged search:** 30 items saved by hand on 2026-10-06 are also in the library. They need a row in this log, or they count as "other methods".
- **First copy:** the first 289 records were also loaded into the personal library "Ma bibliothèque" on the same day. That copy is not the working one.

## Duplicates removed

| Date       | Step                                      | Removed | Remaining           |
|------------|-------------------------------------------|---------|---------------------|
| 2026-10-06 | Same arXiv ID across S0–S3, by script     | 80      | 1,612 arXiv records |
| 2026-10-06 | Same Anthology ID across S0–S3, by script | 9       | 273 ACL records     |

| Date       | Step                                                                                               | Removed | Remaining                            |
|------------|----------------------------------------------------------------------------------------------------|---------|--------------------------------------|
| 2026-10-06 | ACL record with the same title and an author in common with an arXiv record, by script             | 174     | 99 ACL records, 1,711 in total       |
| 2026-10-06 | Same DOI within the OpenAlex run (across strings, plus one DOI indexed twice), by script           | 43      | 1,034 OpenAlex records               |
| 2026-10-06 | OpenAlex record with the same title and an author in common with an arXiv or ACL record, by script | 190     | 844 OpenAlex records, 2,555 in total |

Duplicates whose title changed between preprint and publication are not caught by this rule. They remain to be found in Zotero or Rayyan.

## Pilot screening sample

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Drawn on 2026-10-06 by `scripts/pilot_sample.py` from the 1,711 unique arXiv and ACL records, before the OpenAlex run, at random with seed 20261006: 20 records found by S1, 15 by S2 and 15 by S3. The file to import into Rayyan is `screening/pilot-50.ris`; the keys drawn are in `screening/pilot-50.json`.